

INSPECTION REPORT IR-2026-0147

Equipment: E-2104. Crude / residue heat exchanger, shell side

Unit: Crude Distillation Unit 2

Inspection date: 12-Sep-2026 **Previous inspection:** 20-Sep-2022

Inspector: A. Rao (synthetic)

Thickness readings (UT), interval 4.0 years

CML	Location	t prev (mm)	t actual (mm)	t min (mm)	CR (mm/yr)	RL (yr)
C1	Shell top, inlet end	12.0	11.1	6.2	0.225	21.8
C2	Shell bottom, inlet end	12.0	10.8	6.2	0.300	15.3
C3	Nozzle N3 (outlet)	11.5	9.8	6.2	0.425	8.5
C4	Shell bottom, outlet end	12.0	11.4	6.2	0.150	34.7

Findings

1. Nozzle N3 shows the highest wall loss. Localised pitting up to 0.6 mm deep was found on the tube sheet.
2. No cracking was detected by magnetic particle testing on the shell-to-nozzle welds.

Recommendation

Re-inspect nozzle N3 by UT in 24 months. Review crude desalter performance for chloride carry-over.